

Sequence Recognition of Scene Text Based on CRNN and CTPN Models

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ABSTRACT

Image-based sequence recognition has lately emerged as a prominent study subject in the science of computer vision, while text detection and identification in natural situations has emerged as an active research field. Based on scene text data, this paper addresses the theory of deep learning-based CRNN and CTPN models and the process of processing text. Using CRNN, text recognition can be turned into a time-dependent sequence learning issue, which is commonly employed for indeterminate-length text sequences. Contextual relationships between text images are learned using BLSTM and CTC, thus effectively improving text recognition accuracy and making the model more robust. It also excels in text recognition tests for wordless and lexical-based scenes, as it is not constrained by any predefined language. It produces a more efficient, but smaller, model that is more suited to real-world settings. CRNN recognition accuracy is lower for short texts with large morphological changes, such as artistic words, or texts with large changes in natural scenes. Because of the Anchor setting, CTPN can only detect horizontally distributed text, but a small improvement can detect vertical text by adding horizontal Anchor. As a result of the limitations of the framework, the irregularly inclined text can be detected very broadly.

CCS CONCEPTS

• **Networks** → Network performance evaluation; Network performance analysis.

KEYWORDS

CTPN, LSTM, OCR, Sequence recognition

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1 INTRODUCTION

OCR (Optical Character Recognition) is a process in which an electronic device analyzes characters printed on paper, recognizes their form by detecting color patterns, and then uses character recognition to turn the shape into computer text. Initially, a German scientist named Tauscheck proposed the concept of OCR in 1929. In the early 1960s, the first generation of OCR products appeared. After more than half a century of continuous development, attention was focused on simple open recognition accuracy, and later on new requirements for recognition speed and user-friendliness of the interactive interface. Scene Text Identification (STR) is the recognition of text information in photographs of actual nature scenes. The extraordinarily rich text presentation in natural scene photographs makes text recognition far more challenging than text recognition in scanned document images. Technology for text recognition has advanced quickly in recent years. This is largely because theoretical advancements in pattern recognition, machine learning, computer vision, etc. have encouraged and motivated the development of text recognition technologies. However, there are still many problems to be solved in text recognition. Text recognition technology has both a large number of application needs and a series of theoretical and technical difficulties, which require a lot of R&D efforts to further promote the development of academic research and applications. This study first analyzes and compares the algorithms of CRNN and CTPN. Secondly, the two network structures are implemented through prediction results explore the impact on text recognition in different scenes.

2 METHODOLOGY

2.1 Convolutional Recurrent Neural Network (CRNN)

In 2015, CRNN [1] was proposed to handle image-based sequence recognition challenges, particularly scene text recognition. Its most significant feature is that it does not require a single phrase to be cut at first, but rather transforms text recognition into a sequence-dependent sequence learning problem.

2.1.1 Network Architecture. The CRNN network structure is divided into three components, shown in Fig. 1

Convolutional layers at the bottom of CRNN take features from the input image and generate feature maps automatically. On top of CRNN, a recurrent network is designed to predict each frame of the feature sequence generated by the convolutional layer. The window size of the last two pooling layers among four max-pooling layers is modified from 2×2 to 1×2 , which means that the height of the picture is halved four times, but the width is only halved twice. Because most text graphics are smaller in height and wider in width,

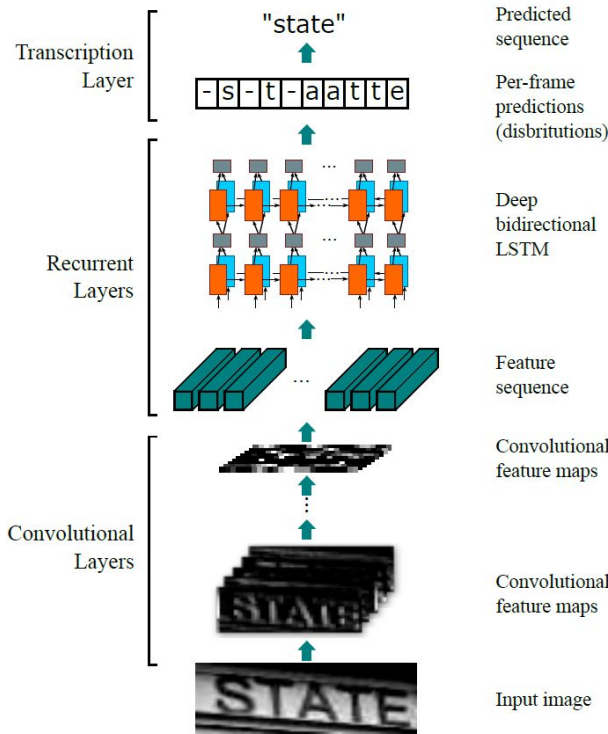


Figure 1: The network architecture [1]

feature maps are rectangular in shape with the same height, breadth, and length. When utilizing a 1×2 pool, the window attempts to preserve widthwise information in order to better recognize English letters (e.g. distinguishing between i and l). CRNN also introduces the Batch Normalization module to speed up model convergence and shorten the training process.

Because transmitting the feature map acquired by CNN into RNN cannot be directly trained, certain changes need to be done to extract the sequence of feature vectors according to the feature map created from the CNN model. The concatenation of all pixels in the column of the feature map yields the corresponding feature vector. And then these feature vectors form a sequence. On the feature map, each feature vector is constructed in columns from left to right, with each column containing 512-dimensional features. Because they are performed on local regions, convolutional layers, max-pooling layers, and activation functions are translation-invariant. As seen in Figure 2, each feature vector specifies a certain width of an image feature.

Since RNNs suffer from the vanishing gradient problem and cannot obtain more contextual information, CRNNs use LSTMs, which are specially designed to capture long-distance relationships. LSTM is unidirectional, which means it only uses past data. The contexts of these two directions are mutually beneficial and complementary in image-based sequences. CRNN makes a bidirectional LSTM by combining two LSTMs, including the former and the latter. Furthermore, bidirectional LSTMs can be layered in multiple layers, and deep structures allow for a higher level of abstraction than shallow structures. A bidirectional LSTM network with two layers of 256 units is shown in Figure 3.

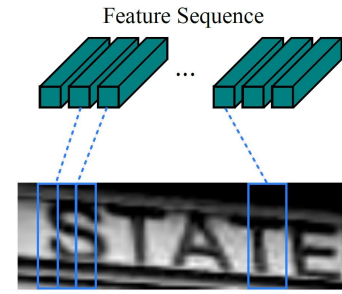


Figure 2: The receptive field [1]

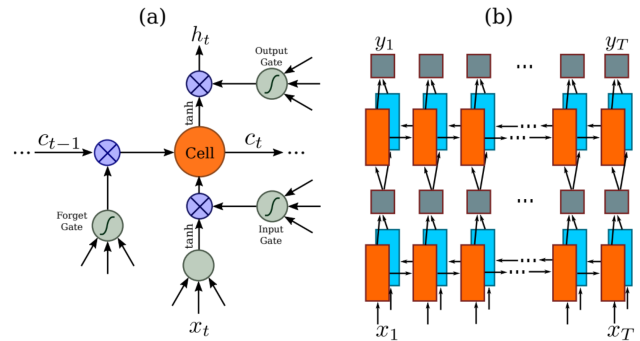


Figure 3: The bidirectional LSTM network [1]

The goal of RNN is to anticipate which character this rectangular region represents, that is, to make predictions and acquire the softmax probability distribution based on the input feature vector. This is a one-dimensional vector containing the number of character classes sent into the CTC layer.

Because each time step comprises an input feature vector x_T and an output probability distribution y_T over all characters, the result is a posterior probability matrix composed of vectors of a particular length, equal to the number of character classes. This posterior probability matrix is passed on to the transcription layer. Although a CRNN is made up of several network designs, it can be trained using a single loss function.

2.1.2 CTC Loss. The transcription layer is in charge of turning the RNN predictions from each feature vector. Transcription is defined as identifying most likely tag sequence combination based on each frame prediction. Dealing with variable-length sequence alignment is an issue in end-to-end OCR recognition. OCR may be modeled as a time-series-dependent text-image problem, and the CTC loss function can be utilized to train the CNN and RNN from start to finish.

The RNN sequence must be converted into the final recognition result. When the RNN does time series classification, there will clearly be a lot of duplicate information. De-redundancy processes, for example, are necessary when a letter is recognized twice in a row.

The "-" symbol is used to denote a blank space. When the network outputs the text prediction, add a "-" between the repeated letters in the sequence. This indicates that blank characters between identical characters will not be combined. Decoding is the process

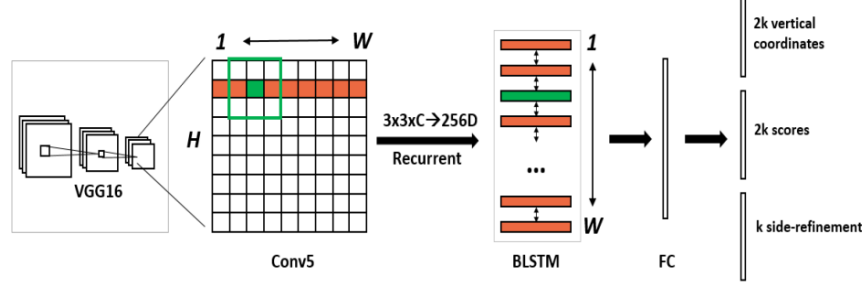


Figure 4: The network architecture [2]

of eliminating consecutively repeated characters from a character sequence, followed by the removal of all "-" characters from the route; encoding is performed by a neural network. We may effectively solve the problem of repeated characters by adding the blank method.

As a result, given the input probability distribution matrix $y = \{y_1, y_2, \dots, y_T\}$, where T is the sequence length, the overall likelihood of mapping to the label text l is:

$$p(l|y) = \sum_{\pi: B(\pi)=1} P(\pi|y) \quad (1)$$

The product of the scores of the relevant characters is used to compute the similarity among each path. It only needs to train the network to maximize this probability value. CTC's loss function is the same as conventional classification's which is a negative maximum likelihood function of probability.

The previous neural network's parameters can be back-propagated by computing the loss function, and the neural network's parameters are updated based on the optimizer used to pick the character corresponding to the most likely pixel area. The CTC does not need to precisely segment the original input character sequence because of this mapping transformation and the total of all possible route probabilities.

2.2 Connection Text Proposal Network (CTPN)

CTPN [2] is an ECCV 2016 text detection method. It is now the most popular and prominent open-source text detection model, capable of detecting horizontal or slightly slanted text lines. In generic object detection, a text line is regarded as a succession among letters rather than a single isolated target. Each character picture on the same text line might have a context. The detection model learns the contextual statistical rules present in the picture during the training phase, which may effectively increase the text block prediction accuracy during the prediction phase. To detect horizontally distributed text in complex settings, CTPN is an efficient algorithm for extracting text in a scene, which can detect the location of text information in a natural environment.

2.2.1 Network Architecture. The original CTPN only recognizes text that is horizontally oriented. The CTPN structure shown in Figure 4 is similar to Faster R-CNN [3], except it includes an LSTM layer. Because the text is continuous, CNN abstract spatial characteristics and sequence features are required for text sequence identification.

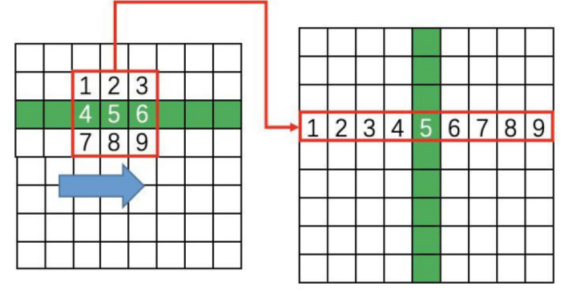


Figure 5: The slicing window [2]

Assuming that N Images are input, the feature is first extracted through the Backbone architecture network VGG16, and the Conv5 layer outputs the feature map of $N \times C \times H \times W$. Since the convolutional network of VGG16 has accumulated 16 Stride through 4 pooling layers. That is, one pixel in the Feature map output by the Conv5 layer corresponds to 16 pixels of the original image. Then, on Conv5, run a sliding window of a specific size shown in Figure 5, coupling each point with the surrounding features to produce a feature vector. The outcome, as seen in the picture 6below, is a feature map which is still the spatial feature learnt by CNN.

Then continue to perform the Reshape operation on the Feature map output in the previous step,

$$\text{Reshape} : N \times 9C \times H \times W \rightarrow (NH) \times W \times 9C$$

The Bi-LSTM is then fed a data stream with $Batch = NH$ and a maximum time duration $Tmax = W$ to learn the sequence characteristics of each row. The output of Bi-LSTM is $(NH) \times W \times 256$, and the form is then recovered using reshape,

$$\text{Reshape} : (NH) \times W \times 256 \rightarrow N \times 256 \times H \times W$$

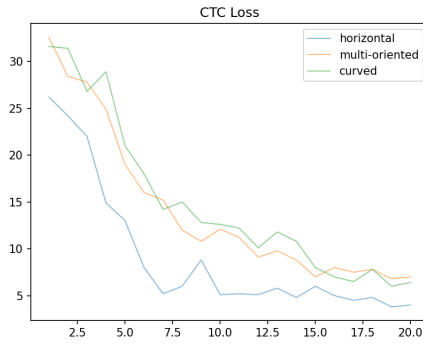
This feature combines spatial and sequence information learnt by Bi-LSTM. It then becomes a feature of $N \times 512 \times H \times W$ through the FC layer. Finally, Text Proposals are acquired after an RPN network comparable to Faster RCNN.

2.2.2 Loss Function. The loss function is divided into 3 parts as follows,

- The problem in anchor foreground and background categorization is that this Loss is used to determine whether or not each Anchor includes text.
- Error in vertical coordinate offset regression: This Loss, like Smooth L1, is used to oversee the learning of each Bouding box

Table 1: Evaluation metrics

Epoch	CTC loss	Learning rate	Sequence distance	Accuracy
1/10	26.673628	0.054214	0.334685	0.467965
2/10	22.656799	0.051827	0.301892	0.548928
3/10	13.536508	0.047127	0.279817	0.591738
4/10	7.376287	0.043622	0.025287	0.623289
5/10	5.756758	0.039267	0.224893	0.689327
6/10	5.961872	0.037821	0.201738	0.713283
7/10	6.284389	0.031738	0.173278	0.701784
8/10	5.183782	0.029176	0.158378	0.749872
9/10	4.876783	0.023648	0.139389	0.771638
10/10	3.998073	0.020972	0.128473	0.781256

**Figure 6: CTC loss values**

regression y-direction offset including an anchor. Where v_j is the Anchor that has text in s_i or a vertical $IoU > 0.5$ with Groud truth. c) Anchor x at the border correction error: This Loss is used to oversee the learning of the x-direction offset of each Anchor holding text in the Bouding box regression, which is the same as the y-direction.

The loss function is shown below,

$$\begin{aligned}
 Loss(s_i, v_j, o_k) = & \frac{1}{N_s} \sum_i L_s^{cls}(s_i, s_i^*) + \frac{\lambda_1}{N_v} \sum_j L_v^{reg}(v_j, v_j^*) \\
 & + \frac{\lambda_2}{N_o} \sum_j L_o^{reg}(o_k, o_k^*)
 \end{aligned} \quad (2)$$

where L_s refers to the classification loss that predicts whether each anchor contains a text area, L_v is the regression loss of the center y coordinate cy and height h of each anchor in the text area, and L_o is the regression loss of the center x coordinate cx of the anchors on both sides of the text area.

3 EXPERIMENTAL VALIDATION AND DISCUSSION

3.1 Datasets

This work used the Total-Text dataset, which is now part of the training set of the biggest curved text dataset - Arbitrary-Shaped Text dataset - to better validate the performance of CRNN and CTPN (ArT). TOTAL-TEXT is a dataset of English curve text at the

word level. It has 1555 distinct visuals and more than three different text orientations, including horizontal, multi-oriented, and curved.

3.2 Comparative Evaluation

CRNN is primarily used to address image-based sequence recognition issues, particularly those involving scene text recognition. Its key advantage is that it can execute end-to-end training on sample data without character segmentation and can identify text sequences of any length. It also has fewer parameters than typical DCNN models and takes up less storage space. It is not restricted to any predetermined thesaurus and has demonstrated outstanding performance in thesaurus-free and thesaurus-based scene text recognition tasks. Figure 6 below depicts the CTC loss curve and a related CRNN accuracy table for three separate scene text recognition scenarios.

As can be observed in Table 1, CRNN's resilience is not particularly good. Because there are not only conventional texts in the horizontal direction in today's natural scene text recognition, but also texts with extreme aspect ratios and curved texts. For such texts, the recognition impact of CRNN is poor.

CTPN is a target detection method-based text detection model. CTPN has a very good effect on horizontally oriented text identification since it enhances Faster RCNN and merges bidirectional LSTM. Another advantage of CTPN is that it changes the text detection job into a succession of small-scale text box detection and then converts it to a more fine-grained task that can produce more accurate text boxes.

One of the most essential aspects of this technique is the use of RNN into text detection, as well as the use of the concept of differentiation to decrease the inaccuracy of the results to be identified, and the use of a fixed-width anchor to identify and segment into Proposal for many blocks. The text section we need to detect is the final combined sequence. The efficient and smooth integration between CNN and RNN enhances accuracy significantly. Table 2 depicts the experimental comparison.

Because of the combination of CNN and bidirectional LSTM, the study findings reveal that CTPN is quite successful for horizontal text detection. Furthermore, side-refinement enhances text box boundary prediction accuracy. It locates text locations using a set of vertical anchors and turns the text detection task into the detection of a succession of small-scale text boxes. CTPN, however, can only

Table 2: Accuracy for CRNN and CTPN

	Horizontal	Multi-oriented	Curved
CRNN	0.736729	0.633252	0.615635
CTPN	0.932172	0.717383	0.672367

Table 3: Results

Image	CRNN	CTPN
	CARMINE'S CARMINES	CARMINE'S CARMINES
	OXFORD STREET W1 CITY OF WESTMINSTER	OXFORD STREET W1 CITY OF WESTMINSTER

recognize horizontally spread text owing to the Anchor setting, and a little enhancement may be made to detect vertical text by adding a horizontal Anchor.

However, due to framework limitations, the detecting impact of irregular oblique text is highly widespread. Furthermore, CTPN incorporates bidirectional LSTM to learn text sequence characteristics, which aids in text detection. However, since the advent of LSTM, it has become quite simple to increase the gradient during training, which must be managed with caution. The prediction results of CRNN and CTPN on scene text are shown in the image shown in Table 3.

In addition, accuracy can be improved by using an end-to-end system for locating and identifying text in natural scene images and text-based image retrieval. The system is based on the region proposal mechanism for detection and deep convolutional neural network for recognition. [4]

4 CONCLUSION

The predicted impact of CRNN and CTPN of scene text recognition is compared in this study. First, CRNN combines the benefits of CNN and RNN; it can receive input pictures of varying sizes and make predictions of varying durations. It works directly with coarse-grained labels (like words) and does not require extensive labeling of each individual element (like letters) during the training phase. Furthermore, because CRNN does not use fully connected layers like traditional networks, it generates a more compact and efficient model. Experiments on benchmark datasets for scene text recognition show that CRNN outperforms conventional techniques and other CNN- and RNN-based algorithms in terms of performance. Because CRNN is a broad framework, it may be used to solve different picture sequence prediction issues.

Furthermore, CTPN employs the anchor regression method, allowing RPN to identify multi-size objects using a single-size sliding

window. CNN and bidirectional LSTM are used to extract picture features, whereas LSTM is used to extract sequence contextual data. CTPN can only recognize text in the horizontal direction, and words in the vertical direction will be disconnected. If this occurs, the connection strategy for managing the anchor must be modified.

Back in 2011, a paper used a system rooted in universal object recognition to study text detection and recognition in natural images. [5] Later, a system was developed to extract text from images. [6] In recent years, an easy-to-implement new brush detector [7], a framework for recognizing natural scene text [8], and a graphical model incorporating conditional random fields (CRF) [9] have been developed. In addition, training a common object degree metric to produce a small window set of candidate objects has been shown to further improve performance to 99.5% of DR. [10]

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